01 /02

Lab Component Sorter: ESP32 + Image Recognition over Wi-Fi

TYPE
Group project, team of 5

DATE
Winter 2025

ROLE
Firmware and pin mapping

STACK

ESP32-CAM | Arduino Nano | Servos | Image recognition | Wi-Fi | 3D printing

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FIG. 1 — CAD: CHUTE, FLAP, BINS

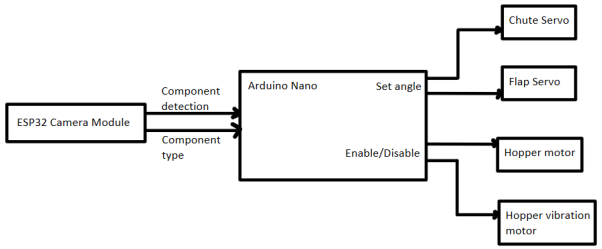
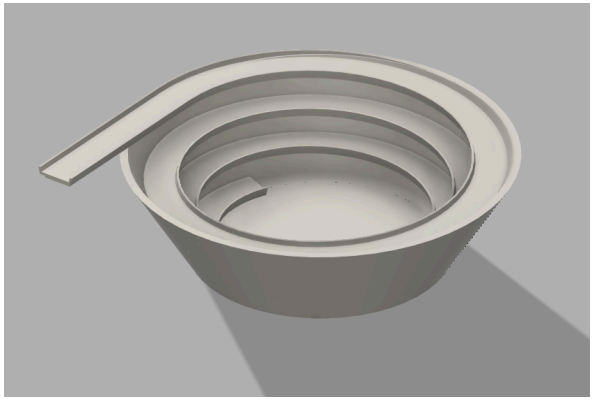


FIG. 2 — SYSTEM DIAGRAM



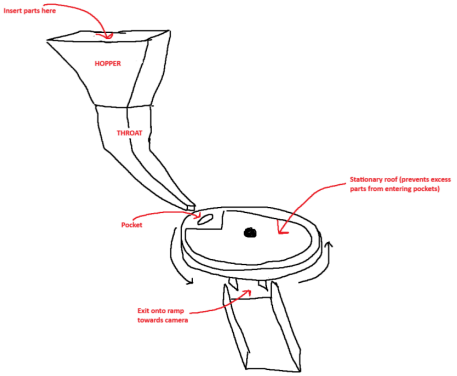


FIG. 3 — ISOLATOR: SPIRAL BOWL + EARLY CONCEPT

02 /02

Optics Lab: Interferometry and Data Analysis in MATLAB

TYPE
Carleton physics lab

ROLE
Setup and analysis, with a partner

STACK

- MATLAB
- Least-squares fitting
- Error propagation
- Formal reporting

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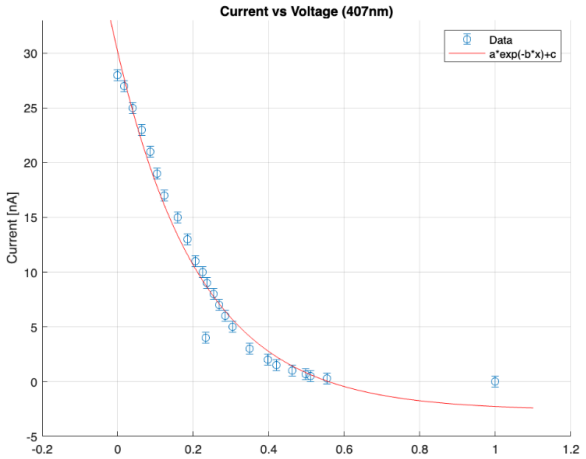


FIG. 1 — NONLINEAR FIT, MATLAB

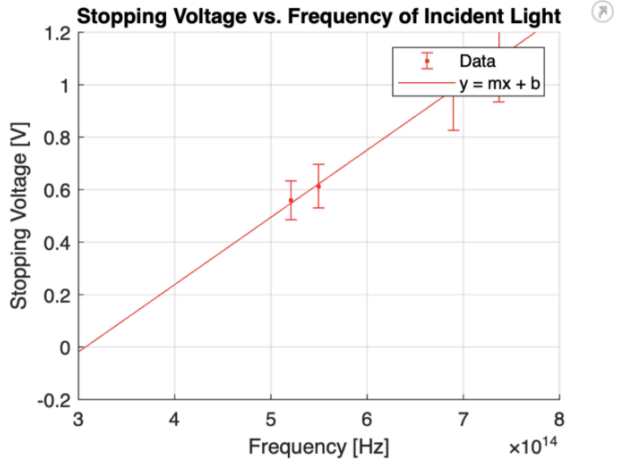


FIG. 2 — LINEAR FIT WITH ERROR BARS

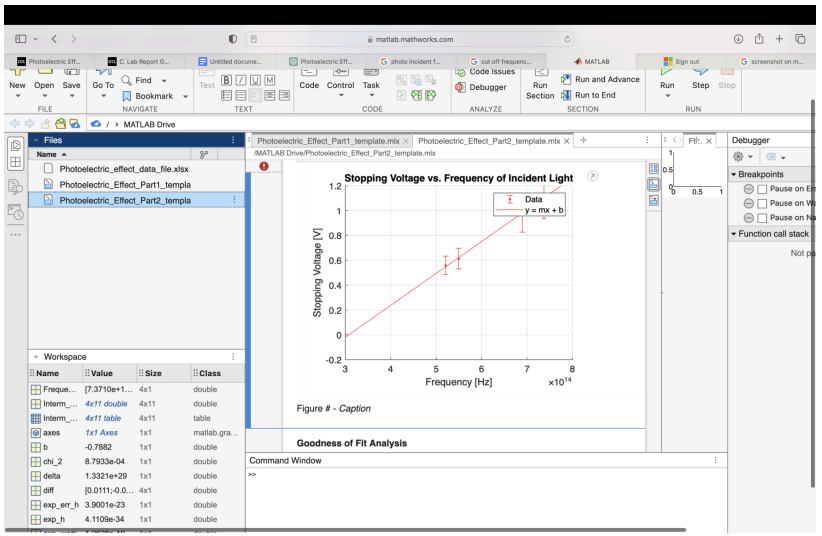


FIG. 3 — MATLAB LIVE SCRIPT AND WORKSPACE